

Francisco José Díaz Villarreal

Software Engineer | Full Stack Developer | Spec Driven Development | Project Leader | Data & Game Specialist
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Software Engineer with PhD in Applied Sciences has solid experience in the design and development of **scalable web applications**, **RESTful APIs**, systems based on data sciences and video games (**Chess-Based Systems**). Specialized in **Spec Driven** and **Full Stack** development, system integration and application of good software engineering practices. Experience working with multiple languages, frameworks, applying **Github** and modern software architecture. Profile with strong analytical skills, technical leadership and focus on real products.

➤ PROFESSIONAL EXPERIENCE

✓ Full Stack Developer & Software Engineering Instructor

Universidad Tecnológica Nacional (UTN) – CUDI | 2023 – 2025

- Design and development of Full Stack web applications using **Vue.js**, **React.js**, **frontend Bootstrap.js** and **Node.js**, **C**, **C++**, **C#** in backend
- Building and integrating **RESTful APIs** connected to **SQL** and **NoSQL** databases
- Applying workflows with **Github**
- Database design **implementation**, system integration and application structure

✓ SQL & Database Developer (Academic Projects)

Universidad Argentina de la Empresa (UADE) | 2021 – 2022

- Design and optimization of **relational database schemas** to ensure information integrity
- Developing and optimizing **SQL Server Management Studio** to improve data performance and efficiency
- Application of **data normalization** principles and good relational modeling practices

✓ Systems & Software Support Engineer

Servicio Autónomo de Registros y Notarías (SAREN) | 2009 – 2019

- Support and maintenance of enterprise information systems used in administrative processes
- Improvement of digital document management processes, contributing to information management
- Implementation of technical solutions to optimize system reliability and operational continuity

✓ Project Lead – Masterdrez Science Video Game

Centro Nacional de Desarrollo e Investigación en Tecnología (CENDITEL) | 2008 - 2012

- Coordinator responsible for and technical management of the development of the video game Masterdrez
- Concept Author, Software Architecture Definition, Game System Logic and Design
- Supervision, implementation, technical direction of a team of engineers with master's degree

✓ Software Development Instructor & Technical Mentor

Universidad Bolivariana de Venezuela (UBV) | 2004 – 2010

- Training in programming, software engineering and computer architecture
- Mentoring teams in software projects, game development, promoting development
- Technical coordination of applied development projects aimed at building solutions

✓ Computer Science & Chess Systems Instructor | Strategic Thinking & Game Logic

Unidad Educativa “Canónigo Uzcátegui” | 2004 - 2009

- Taught computer fundamentals, logic, and basic programming structures
- I applied chess concepts as a model of strategic systems and decision-making
- Coordinated educational initiatives at the municipal level related to health and cognitive development
- Led the implementation of the National School Chess Program

➤ **ACADEMIC BACKGROUND**

- ✓ **Universidad de los Andes (ULA)**
PhD in Applied Science – [Degree](#) – (2024)
- ✓ **Instituto Superior Politécnico José Antonio Echeverría (ISPJAE)**
M. Sc. in Applied Computer Science – [Degree](#) – (2010)
- ✓ **Universidad de Buenos Aires (UBA)**
Diploma in Initialization in Programming and Data Analysis (Python) – [Degree](#) – (2025)
- ✓ **Universidad Valle del Momboy (UVM)**
Computer Engineering – [Degree](#) – (2004)
- ✓ **Centro de Formación Profesional (CFP)** - Vocational Technical Education Level: 3
Professional Specialization in Video Game Programming (C#) – [Certificate](#) – (2022)
Programmer Track (TypeScript/SQL) – [Certificate](#) – (2021)

➤ **COMPUTER COURSES**

- **Adaptive Behavior** – [Certificate](#)
- **Manual, Agile, and Tooled Testing** – [Certified](#)

➤ **COMPUTER CERTIFICATIONS**

- **Database Management** (Python) – [Certificate](#)
- **Lauching Into Machine Learning** (Python) – [Certificate](#)
- **Full Stack Python** (HTML, CSS, JavaScript, MySQL, Vue, Bootstrap) – [Certified](#)
- **Full Stack .NET** (HTML, CSS, JavaScript, MySQL, Bootstrap, Node.js, Handlebars) – [Certified](#)
- **Full Stack Java** (HTML, CSS, JavaScript, MySQL, Vue, Bootstrap) – [Certified](#)
- **Spring Specialization** (Java) – [Certificate](#)
- **React JS** (JavaScript) – [Certificate](#)
- **Salesforce Platform** (Apex) – [Certified](#)
- **Unity 2D** (C#) – [Certified](#)
- **IBM Cloud** (Kubernetes) – [Certified](#)

➤ **LANGUAGE**

- **Certificate of Proficiency at Instrumental Level of English:** in Engineering - [Certificate](#)

➤ **PERSONAL CAPABILITIES**

- **Employability Skills** – [Certificate](#)
- **Soft Skills** – [Certificate](#)

➤ **TECHNICAL SKILLS**

- **Software Architecture:** MVC, MVVM, Layered Architecture, Microservices, API Integration
- **Game Development:** Unity (C#), C++, JavaScript, implementación de lógica de IA en juegos
- **Backend Development:** Python, Java (Spring Boot), Node.js, .NET, PHP, RESTful APIs
- **Frontend Development:** React.js, Vue.js, JavaScript, TypeScript, HTML5, CSS3
- **Bases de Datos:** MySQL, SQL Server Management Studio, MongoDB, SOQL y SOSL
- **AI & Data:** Python, TensorFlow, Machine Learning, Data Analysis, OpenCode, Codex
- **Cloud & Platforms:** IBM Cloud, Salesforce Platform
- **Testing:** Fundamentos de Unit Testing
- **DevOps & Tools:** Git, GitHub, Trello
- **Responsive Design:** Bootstrap.js